

AROOJ FATIMA

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EDUCATION

National University of Sciences & Technology (NUST)

Bachelor of Science in Bioinformatics

CGPA: 3.41/4.00

Islamabad, Pakistan

Sep 2023–Jun 2027

Relevant Coursework: Discrete Mathematics, Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Computer Organization and Assembly Language, Parallel & Distributed Computing, Linear Algebra, Probability and Statistics, Software Engineering, Fundamental of Computer Programming, Object Oriented Programming, Database Systems

RESEARCH INTERESTS

Theoretical Computer Science, Formal Verification, Interactive Theorem Proving, Systems Software Verification, C Program Verification, Programming Languages and Formal Semantics, and Verified Compilation.

RESEARCH & PROFESSIONAL EXPERIENCE

Formal Verification Engineer Intern

Formal Land, Paris, France / Remote

Jun 2026–Present

[rocq-of-rust](#)[Public contributions](#)

- Updated rocq-of-rust for a newer version of Revm (a Rust implementation of the Ethereum Virtual Machine), adapting its Rust-to-Rocq translations, functional specifications, and proofs to changes in Revm’s architecture, and added evaluation through OCaml extraction.
- Currently developing instruction-level correspondence proofs between Revm and go-ethereum in Rocq, relating their models of stack, memory, storage, gas, host state, control flow, and errors. Also built mutation tests and runners for official Ethereum test cases.

Student Researcher

System Analysis and Verification Lab, NUST

Jun 2025–Present

[Playground](#)[Algorithms](#)

- Explored Lean 4 through game-based exercises and small formalizations, then moved to Rocq and its functional programming language, Gallina, for larger verification projects.
- Proved correctness properties of BST insertion, insertion sort, merging sorted lists, and MPI block-decomposition formulas, and formalized red-black-tree invariants and repair operations.
- Currently working on:** Formal Verification of μC /OS-III kernel in Rocq and Verified Software Toolchain (VST).

Summer Research Intern

Deep Learning Lab, NCAI, NUST

Jun–Aug 2025

[Repository](#)[Certificate](#)

- Trained models across decision trees, neural networks, ResNet50, and YOLO on CIFAR and custom datasets
- Reproduced active-learning research for medical-image analysis and compared selection methods using TF-IDF and Sentence-BERT representations.

ACCEPTED RESEARCH

- [1] **Arooj Fatima and Saad Saghir.** “Towards Machine-Checked Verification of the μC /OS-III Kernel Using Rocq and the Verified Software Toolchain.” 2026

Verified with VST four C priority-bitmap functions for a pinned 32-priority μC /OS-III configuration. Generated their CompCert Clight representation using `clightgen` and proved initialization, insertion, removal, and highest-ready-priority lookup correct against abstract ready-priority specifications.

Presented at the **FMCAD 2026 Student Forum.**

[Forum Listing](#)[Repository](#)[Poster](#)

- [2] **Arooj Fatima, Saad Saghir and Adnan Rashid.** “A Comparative Formal Verification Framework for General-Purpose CPU Scheduling in Rocq.” 2026

Developed Rocq models and proofs for six CPU schedulers, including red-black-tree-based CFS and identified reusable proof patterns for minimum keys, layered priorities, and ticket intervals.

Accepted for publication at the **6th IEEE International Conference on Digital Futures and Transformative Technologies (ICoDT2 2026).**

[Repository](#)

OPEN-SOURCE CONTRIBUTIONS

Contributor to coqutil and Parseque

Jul 2026–Present

Rocq

[coqutil](#) and [Parseque](#)

- Authored seven merged pull requests adding build documentation, reusable characterization lemmas for lists, options, results, and length-indexed tuples, and multiline parser-position tests.
- Contributions: [#168](#), [#169](#), [#170](#), [#171](#), [#176](#), [#178](#), and [#22](#).

PROJECTS & SELF-LEARNING

Software Foundations

Dec 2025–Present

Rocq / Self-learning

Private per authors' request

Working through selected exercises from *Logical Foundations* and *Programming Language Foundations*, covering inductive types, recursion, pattern matching, structural induction, higher-order functions, inductive relations, big-step operational semantics, program equivalence, semantics-preserving optimizations, nondeterminism, and Hoare logic.

Verified Distributed Protein-Network Analysis

May 2026

PySpark, Rocq, and STRING / Part of Parallel & Distributed Computing Course

[Repo](#)

Processed 13.7M protein interactions into 473,860 high-confidence edges using PySpark; formalized and verified Large-Star and Small-Star at a fixpoint in Rocq.

Biological Sequence Formalization

Jan 2026

Rocq / Self-learning

[Repo](#)

Formalized DNA/RNA sequence operations and proved properties of complementation, pairing, concatenation, repetition, reversal, and GC-content preservation.

Concurrent Web-Server Simulation on FreeRTOS

Dec 2025

C, FreeRTOS, RISC-V, and QEMU / Part of Operating Systems Course

[Repo](#)

Built a concurrent web-server simulation demonstrating priority-based scheduling, producer-consumer task queues, and mutex-based synchronization for HTTP requests.

Parallel Algorithm Performance Analysis

Nov 2025

C, OpenMP, Pthreads, and MPI / Part of Operating Systems Course

[Repo](#)

Implemented and benchmarked parallel matrix multiplication and merge sort on datasets ranging from 10 to 10 million elements, comparing the overhead and scaling behavior of three parallel-programming models.

Operating Systems Concepts | OSTEP

Nov 2025

C / Self-learning

[Repo](#)

Implemented OSTEP exercises on process API (fork/exec/wait), CPU scheduling algorithms, and memory virtualization.

Standard ML and HOL4

Jul 2025

Standard ML (SML), HOL4 / Self-learning

[Repo](#)

Studied Standard ML and HOL4 through exercises involving recursion, currying, HOL terms and types, algebraic datatypes, and termination-checked function definitions.

OTHER ACTIVITIES

Curriculum Design and Mentorship Team

Oct 2025–Present

SPARK

Technical Writing and Research Resource Curation

Jan 2025–Mar 2026

Published technical notes on TCS/FV/QC and curated research opportunities.

[Notion](#)

[Medium](#)

[Repo](#)

Event Organization and Administration

Oct 2024–Apr 2025

NUST Entrepreneurs Club / ICoDT2 2024 Conference

[Certificate](#)

[Certificate](#)

AWARD

FMCAD 2026 Student Forum Travel Award, Selected recipient

2026

TECHNICAL SKILLS

Verification and Functional Programming: Rocq, Gallina, Lean 4, OCaml, Verified Software Toolchain (VST), CompCert/Clight, SML, HOL4

Programming, Systems and Tools: Rust, C, Python, Git, Linux CLI, FreeRTOS, MPI, OpenMP, L^AT_EX